```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\backend> docker build -t khangtrong/notes-backend:v1 .
[+] Building 23.8s (10/10) FINISHED                                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                        0.1s
 => => transferring dockerfile: 647B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:21-jdk                                                  3.5s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                     0.0s
 => [internal] load .dockerignore                                                                                           0.0s
 => => transferring context: 2B                                                                                             0.0s
 => [1/4] FROM docker.io/library/eclipse-temurin:21-jdk@sha256:e8d57b6c5c73d93212a4384f588309b44bc75fd15b38f78e6a6db1b350dc4ef  17.7s
 => => resolve docker.io/library/eclipse-temurin:21-jdk@sha256:e8d57b6c5c73d93212a4384f588309b44bc75fd15b38f78e6a6db1b350dc4ef3  0.0s
 => => sha256:a3307ab29355ce71f400857b35cca7442414151b6d8d767171d67282f986ad07 157.84MB / 157.84MB                         16.1s
 => => sha256:24ebe7eabaf26192ec8e30edade0a6df98ef2590420dfb0ef843456388a54147 158B / 158B                                  1.0s
 => => sha256:df8fd45c97cdc03c294a3813fd5dbdfccd2e3609de14ffa984b6f160afbb7edf 2.28kB / 2.28kB                               1.4s
 => => sha256:a3629ac5b9f4680dc2032439ff2354e73b06aecc2e68f0035a2d7c001c8b4114 29.73MB / 29.73MB                             9.2s
 => => sha256:220e5f46c2556ad3d4e13b9f9820a6a493d998f1bc423515840a27d7a4f41ffe 22.96MB / 22.96MB                             9.2s
 => => extracting sha256:a3629ac5b9f4680dc2032439ff2354e73b06aecc2e68f0035a2d7c001c8b4114                                   0.9s
 => => extracting sha256:220e5f46c2556ad3d4e13b9f9820a6a493d998f1bc423515840a27d7a4f41ffe                                   0.6s
 => => extracting sha256:a3307ab29355ce71f400857b35cca7442414151b6d8d767171d67282f986ad07                                   1.4s
 => => extracting sha256:24ebe7eabaf26192ec8e30edade0a6df98ef2590420dfb0ef843456388a54147                                   0.0s
 => => extracting sha256:df8fd45c97cdc03c294a3813fd5dbdfccd2e3609de14ffa984b6f160afbb7edf                                   0.0s
 => [internal] load build context                                                                                           1.7s
 => => transferring context: 46.80MB                                                                                        1.6s
 => [2/4] WORKDIR /app                                                                                                      0.3s
 => [3/4] COPY target/*.jar app.jar                                                                                         0.1s
 => [4/4] RUN mkdir -p /app/uploads                                                                                         0.3s
 => exporting to image                                                                                                      1.7s
 => => exporting layers                                                                                                     1.3s
 => => exporting manifest sha256:7a273763cf000599f00b16fc0c37ade2c6f212a8c1a289285f356df9c1d617f0                           0.0s
 => => exporting config sha256:0edc59d0aa073235a82b99c5f8eac1b2de92c3428c525074e8bc239a9064a2e5                             0.0s
 => => exporting attestation manifest sha256:a2242952acc5fd440bec9232cc816b7b2e56c4f8278c937cf014c9c9061fa1d2               0.0s
 => => exporting manifest list sha256:f9a45cd59a216ffa2ea59c453a5ca0fb06926ff556fa5814cbc8b3963f0dad4c                      0.0s
 => => naming to docker.io/khangtrong/notes-backend:v1                                                                      0.0s
 => => unpacking to docker.io/khangtrong/notes-backend:v1                                                                   0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/k2gerb2w423kr0fh451w1zr3k
```

Image 1. Build backend image

```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\backend> docker push khangtrong/notes-backend:v1
The push refers to repository [docker.io/khangtrong/notes-backend]
bfa16ba7375f: Pushed
24ebe7eabaf2: Pushed
bb3c6ac5521c: Pushed
220e5f46c255: Pushed
df8fd45c97cd: Pushed
a3629ac5b9f4: Pushed
a6c8bf93b631: Pushed
a144dea7b648: Pushed
a3307ab29355: Pushed
v1: digest: sha256:f9a45cd59a216ffa2ea59c453a5ca0fb06926ff556fa5814cbc8b3963f0dad4c size: 856
```

Image 2. Push backend image to docker hub

```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\frontend> docker build -t khangtrong/notes-frontend:v1 .
[+] Building 317.8s (16/16) FINISHED                                                                         docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                        0.1s
 => => transferring dockerfile: 863B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                             3.5s
 => [internal] load metadata for docker.io/library/node:20-alpine                                                           3.5s
 => [auth] library/node:pull token for registry-1.docker.io                                                                 0.0s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                                0.0s
 => [internal] load .dockerignore                                                                                           0.0s
 => => transferring context: 2B                                                                                             0.0s
 => [build 1/6] FROM docker.io/library/node:20-alpine@sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b6081d2f60fd7d5d430e9c  9.1s
 => => resolve docker.io/library/node:20-alpine@sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b6081d2f60fd7d5d430e9c        0.0s
 => => sha256:3dcec91425079e7b455efc5f2a18d026450c47c9382c41897620afc6b1424e44 1.26MB / 1.26MB                              0.8s
 => => sha256:41b3afaea3b1b1ab04a268431e10dcace7883019a5da7d326aa35dc9713fcbb5 445B / 445B                                  1.0s
 => => sha256:c2b4197efb6ccd7f8b482ae7800f1c9c78c044ea1925878873000 80bcff6b2c9 42.78MB / 42.78MB                            4.7s
 => => sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB                              1.9s
 => => extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612                                   0.3s
 => => extracting sha256:c2b4197efb6ccd7f8b482ae7800f1c9c78c044ea1925878873000 80bcff6b2c9                                  3.8s
 => => extracting sha256:3dcec91425079e7b455efc5f2a18d026450c47c9382c41897620afc6b1424e44                                   0.2s
 => => extracting sha256:41b3afaea3b1b1ab04a268431e10dcace7883019a5da7d326aa35dc9713fcbb5                                   0.1s
 => [internal] load build context                                                                                           262.4s
 => => transferring context: 313.27MB                                                                                       262.3s
 => [stage-1 1/2] FROM docker.io/library/nginx:alpine@sha256:b0f7830b6bfaa1258f45d94c240ab668ced1b3651c8a222aefe6683447c7bf55  6.9s
 => => resolve docker.io/library/nginx:alpine@sha256:b0f7830b6bfaa1258f45d94c240ab668ced1b3651c8a222aefe6683447c7bf55          0.0s
 => => sha256:e096540205d5d1ab2c2647d716491020d81e3f167a33606665b193d75540857f 20.18MB / 20.18MB                            2.9s
 => => sha256:0abf9e5672665202e79f26f23ef5dbd12558e2ea51ac32807922ab76fdb24ab0 1.40kB / 1.40kB                              1.0s
 => => sha256:085c5e5aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06 1.21kB / 1.21kB                              0.7s
 => => sha256:33f95a0f3229b49e777082e801b882b13fcc5b4e389410ce8eb066f4d58c71b9 403B / 403B                                  0.9s
 => => sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d 953B / 953B                                  0.8s
 => => sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a 626B / 626B                                  0.7s
 => => sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748fcf73f3c8b2d10816ae3 1.86MB / 1.86MB                              0.9s
 => => extracting sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748fcf73f3c8b2d10816ae3                                   0.3s
 => => extracting sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a                                   0.1s
 => => extracting sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d                                   0.0s
 => => extracting sha256:33f95a0f3229b49e777082e801b882b13fcc5b4e389410ce8eb066f4d58c71b9                                   0.1s
 => => extracting sha256:085c5e5aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06                                   0.1s
 => => extracting sha256:0abf9e5672665202e79f26f23ef5dbd12558e2ea51ac32807922ab76fdb24ab0                                   0.1s
 => => extracting sha256:e096540205d5d1ab2c2647d716491020d81e3f167a33606665b193d75540857f                                   1.6s
 => [build 2/6] WORKDIR /app                                                                                                0.6s
```

Image 3. Build frontend image

```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\frontend> docker push khangtrong/notes-frontend:v1
The push refers to repository [docker.io/khangtrong/notes-frontend]
0abf9e567266: Waiting
da7c973d8b92: Pushed
7b3eec2d0d05: Pushed
1074353eec0d: Pushed
25f453064fd3: Pushed
567f84da6fbd: Pushed
e096540205d5: Pushed
29943ec2d465: Pushed
33f95a0f3229: Pushed
085c5e5aaa8e: Pushed
v1: digest: sha256:36247f49a388fec8f4f1d7d9b60039b5a52984041686cacf06aed26277617cba size: 856
```

Image 4. Push frontend image to docker hub

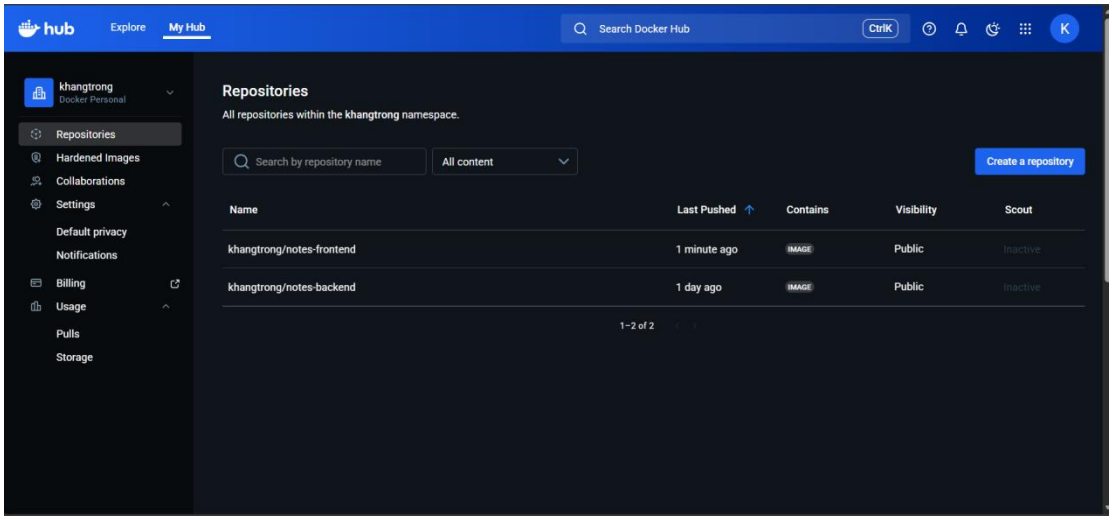

Image 5. List of image on docker hub

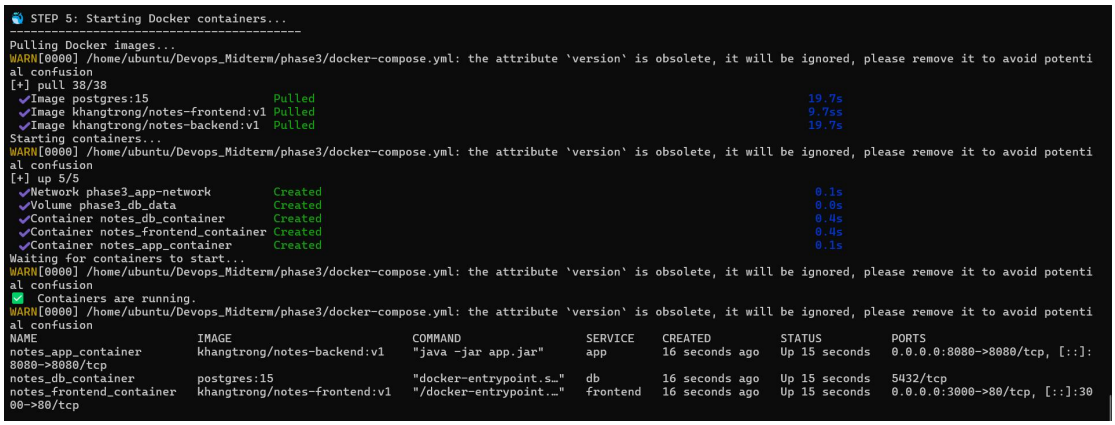

Image 6. Docker containers started using Docker Compose

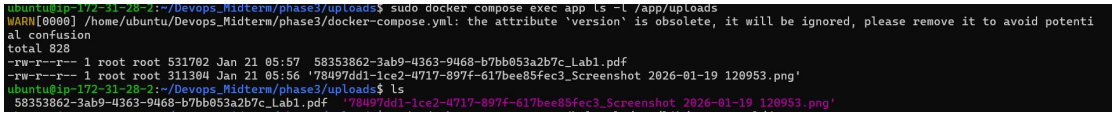

Image 7. Data persistence in a Docker volume

```
 id |                                attachment_url                                | content |         created_at         |            file_name
   |    file_type     | title |         updated_at
----+------------------------------------------------------------------------------+---------+----------------------------+------------------------------
---+-----------------+-------+----------------------------
  1 | /uploads/78497dd1-1ce2-4717-897f-617bee85fec3_Screenshot 2026-01-19 120953.png | 1       | 2026-01-21 05:56:24.841544 | Screenshot 2026-01-19 120953.p
ng | image/png       | 1     | 2026-01-21 05:56:24.841821
  2 | /uploads/58353862-3ab9-4363-9468-b7bb053a2b7c_Lab1.pdf                        | 2       | 2026-01-21 05:57:22.840608 | Lab1.pdf
   | application/pdf | 2     | 2026-01-21 05:57:22.840648
(2 rows)

(END)
```

```
ubuntu@ip-172-31-5-152:~/Devops_Midterm/phase3/uploads$ sudo docker compose ps
WARN[0000] /home/ubuntu/Devops_Midterm/phase3/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potenti
al confusion
NAME                    IMAGE                       COMMAND                  SERVICE    CREATED          STATUS          PORTS
notes_app_container     khangtrong/notes-backend:v1 "java -jar app.jar"      app        10 minutes ago   Up 10 minutes   0.0.0.0:8080->8080/tcp, [::]:
8080->8080/tcp
notes_db_container      postgres:15                 "docker-entrypoint.s…"   db         10 minutes ago   Up 10 minutes   5432/tcp
notes_frontend_container khangtrong/notes-frontend:v1 "/docker-entrypoint.…"  frontend   10 minutes ago   Up 10 minutes   0.0.0.0:3000->80/tcp, [::]:30
00->80/tcp
ubuntu@ip-172-31-5-152:~/Devops_Midterm/phase3/uploads$
```